MIT Art Design and Technology University

MIT School of Computing, Pune

Department Electrical & Electronics Engineering

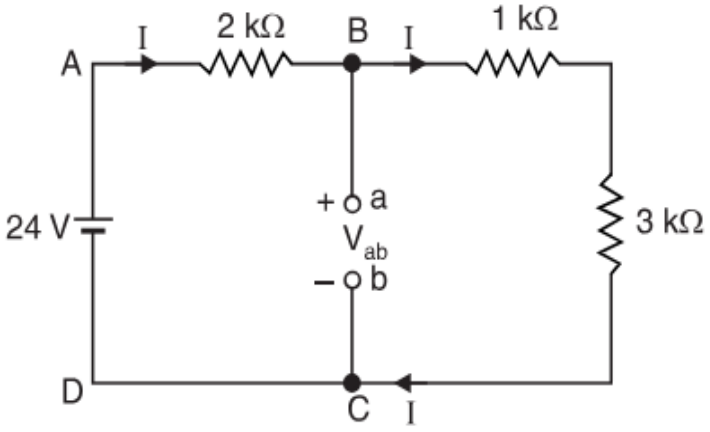
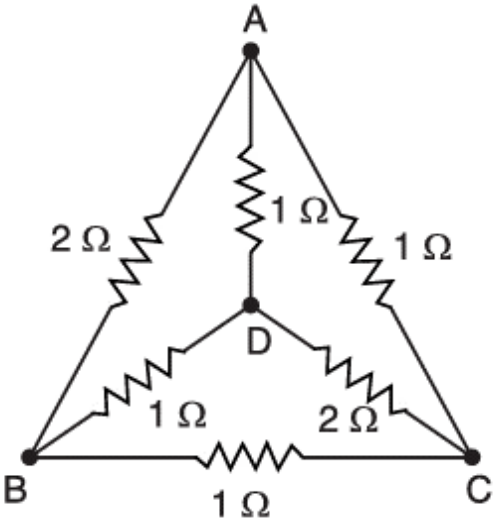
First Year B. Tech

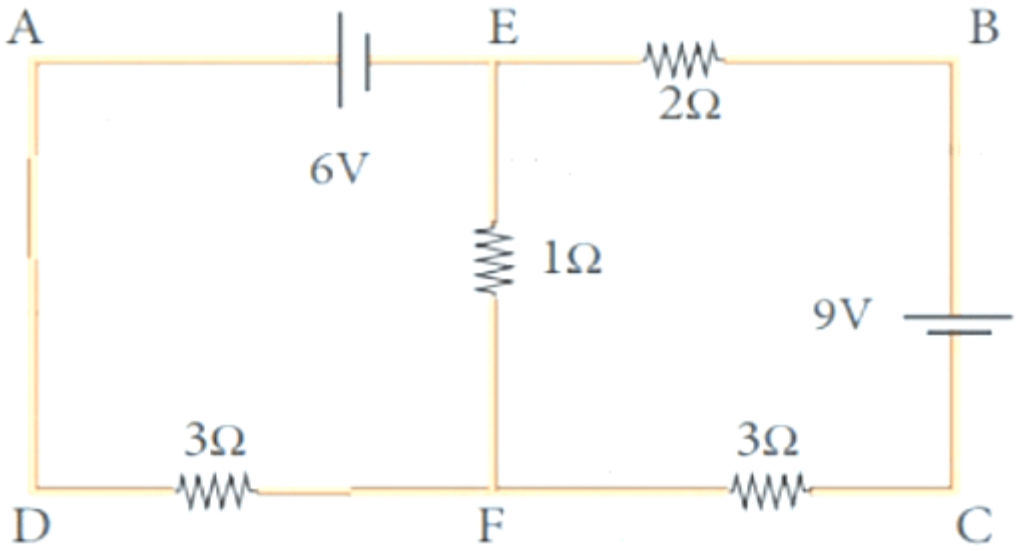
Academic Year 2026-2027 (SEM-I)

Subject: Electrical & Electronics Engineering (23CSE1104)

Theory Assignment-1

1	State & explain with a suitable example, the Superposition theorem as applied in DC resistive circuits
2	State Thevenin's Theorem & draw Thevenin's equivalent circuit.
3	State the following terms: Ohm's Law, Node, Mesh, Branch
4	State Kirchoff's Current law as applied to electrical circuits & explain with a suitable example
5	State Kirchoff's Voltage law as applied to electrical circuits & explain with a suitable example
6	Classify the different types of Electrical networks.
7	Discuss the concept of phase and phase difference along with phasor diagram.
8	With reference to AC quantities, define the following terms: i) waveform ii) Cycle iii) instantaneous value iv) Frequency v) Time period vi) Amplitude
9	State the following terms along with formula. i) Average value, ii) RMS value, iii) Form factor, iv) Peak factor
10	Define Peak factor & form factor. State their values for a sinusoidal quantity.
11	Define Linear & Non-linear networks.
12	Define Unilateral &Bilateral networks with examples
13	Explain Active & Passive networks with diagram.
14	State the formulae to convert the star connected network into its equivalent delta connected network.
15	State the formulae to convert the Delta connected network into its equivalent Star connected network.
16	Write Steps to Analyze the mesh analysis technique
17	Distinguish between Electrical & Electronics Engineering
18	Derive the expression for instantaneous current when voltage $v = V_m sin\omega t$ is applied to a circuit consisting of pure capacitance of 'C' farad. Sketch the

	waveforms & draw phasor diagram.
19	Prove that current in purely resistive circuit is in phase with the applied AC voltage and current in purely capacitive circuit lead applied AC voltage by 90o.
20	Derive the expression for instantaneous current when voltage $v = V_m sin\omega t$ is applied to a circuit consisting of pure inductance of 'L' farad. Sketch the waveforms & draw phasor diagram.
21	Elaborate the concept of series and parallel combination of resistances.
22	Use Kirchhoff's voltage law to find the voltage Vab in Figure. 
23	In the network shown in Figure, find the resistance between terminals B and C using star/delta transformation. 
24	A. An alternating current of frequency 60 Hz has a maximum value of 12 Amp. 1. Write down the equation for instantaneous value. 2. Find the value of current after 1/360 second. 3. Time taken to reach 9.6 Amp. for the first time. B. The waveform of a voltage has form factor of 1.15 and peak factor of 1.5. If the maximum value of voltage is 4500 volts. Calculate average value and rms value.

25	Calculate the current that flows in the 1 Ω resistor in the following circuit using branch current method (KVL), mesh analysis & node analysis, Superposition theorem & Thevenin's theorem. 

Last Date to submit the assignment is 9th September, 2026